PRODUCT USER MANUAL

For Multi Observation Global Ocean 3D Temperature Salinity Heights Geostrophic Currents and MLD Product

MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012

Issue: 2.0

Contributors : N.Verbrugge

Approval Date: November 2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	15/09/20	All	Initial version of the document Created from CMEMS-MOB-PUM-015-001 and CMEMS-MOB-PUM-015-002	S. Guinehut	C. Derval
1.1	15/08/21	I.1, II.1, II.4.2	Minor changes to specify the update frequency and the temporal coverage of rep datasets	S. Guinehut	C. Derval
1.2	12/06/2023	I.2	Update template SSS Upstream evolutions	H. Etienne, N. Verbrugge	Copernicus Marine Product Management
2.0	28/05/2025	All	New dataset for climatological uncertainties for T & S Datasets name change Spatio-temporal resolution change (1/4° to 1/8° and weekly to daily)	N. Verbrugge	H.Etienne

TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DOWNLOAD A PRODUCT ..... Erreur ! Signet non défini.

1) INTRODUCTION **6**

a) Summary 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **7**

a) General Information 7

b) Production System Description 9

c) Dataset details 9

d) Catalogue updating 10

e) Time averaging 10

f) Product use and impact 10

3) FILE FORMAT **11**

4) FILES NOMENCLATURE **11**

5) FILE SIZE **12**

6) STRUCTURE AND SEMANTIC OF NetCDF MAP FILES **12**

7) REFERENCES **13**

8) ANNEX **14**

a) Structure and semantic of NetCDF maps files..... **Erreur ! Signet non défini.**

GLOSSARY AND ABBREVIATIONS

CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Environment Monitoring Service
GLO	Global
MULTIOBS	Multi Observation
NetCDF	Network Common Data Form
PUM	Product User Manual
QUID	Quality Information Document
MLD	Mixed Layer Depth
NRT	Near-Real-Time
REP	REProcessed
SST	Sea Surface Temperature
SSS	Sea Surface Salinity
MDT	Mean Dynamic Topography
SLA	Sea Level Anomaly
WOA	World Ocean Atlas
MFC	Monitoring Forecasting System

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This is the Product User Manual describing the external product **MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012**, developed by CLS Production Unit (MULTIOBS-CLS-TOULOUSE-FR), which provides global Level-4 (L4) analyses of 3D temperature, salinity, geopotential height and geostrophic current from the surface down to 5500-m depth and 2D mixed layer depth (MLD) on a 1/8° regular grid. The product is obtained through a combination of satellite and in-situ data. It includes 5 datasets:

- cmems_obs-mob_glo_phy_nrt_0.125deg_P1D-m , which provides near-real-time (NRT) daily data
- cmems_obs-mob_glo_phy_nrt_0.125deg_P1M-m, which provides near-real-time (NRT) monthly data
- cmems_obs-mob_glo_phy_my_0.125deg_P1D-m, which provides multi-year reprocessed (REP) daily data
- cmems_obs-mob_glo_phy_my_0.125deg_P1M-m, which provides multi-year reprocessed (REP) monthly data
- cmems_obs-mob_glo_phy_mynrt_0.125deg-climatology-uncertainty_P1M-m, which provides monthly climatology of uncertainty for the daily ARMOR3D temperature and salinity fields (REP and NRT).

REP datasets are extended in time once a year, so that they cover 1993 up to 12 to 24 months before present. The REP datasets benefit from the highest quality ocean observations inputs and are produced with a unique system (no change in time in the setup, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.
The manual covers: how the product was created, its content, and which data services are available to access it.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Table 1: MOB-TAC MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012

Product Lines	**MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012**
Geographical coverage	Global: 82.1875°S - 89.9375°N, 179.9375°W - 179.9375°E
Variables (standard_name)	• sea_water_temperature • sea_water_salinity • geopotential_height • geostrophic_eastward_sea_water_velocity • geostrophic_northward_sea_water_velocity • ocean_mixed_layer_thickness • sea_water_temperature_uncertainty • sea_water_salinity_uncertainty
Product Type	Multi Year and Near Real Time
Available time series	cmems_obs-mob_glo_phy_nrt_0.125deg*: 01/2023 – ongoing cmems_obs-mob_glo_phy_my_0.125deg*: 01/1993 to 12 to 24 months before present (see Copernicus Marine Service online catalogue, temporal extension at least once a year)
Temporal resolution	Daily field Average Monthly field, nominally associated with the 15th of the month Monthly climatology for the uncertainties
Target delivery time	Near Real Time: at 16:00 UTC

Product Lines	**MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012**
Delivery mechanism	After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download. Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.
Horizontal resolution	1/8° on a regular grid
Number of vertical levels	50 levels from 0 to 5500-m depth: 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 80, 90, 100, 125, 150, 175, 200, 225, 250, 275, 300, 350, 400, 450, 500, 550, 600, 700, 800, 900, 1000, 1100, 1200, 1300, 1400, 1500, 1750, 2000, 2500, 3000, 3500, 4000, 4500, 5000, 5500; Only up to 1500m for the uncertainties (41 levels)
Format	netCDF-4

b) Production System Description

The methodology used to calculate the product and the variables are described in the QUID documentation https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-012.pdf

c) Dataset details

The NRT datasets cover the period from year-2 to present.
The multi-year datasets cover the period from January 1993 to 12 to 24 months before present.
The monthly static uncertainty dataset is computed over the full multi-year reprocessing period.

Table 2: list of the datasets and variable names and unit for the MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 product: dynamical datasets

MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012
Dataset: **cmems_obs-mob_glo_phy_nrt_0.125deg_P1D-m,** **cmems_obs-mob_glo_phy_nrt_0.125deg_P1M-m,** **cmems_obs-mob_glo_phy_my_0.125deg_P1D-m,** **cmems_obs-mob_glo_phy_my_0.125deg_P1M-m** **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
to (degree Celsius)**:** temperature sea_water_temperature
so (practical salinity unit)**:** salinity sea_water_salinity
zo (meter): absolute height geopotential_height
ugo (m s-1): geostrophic zonal velocity from thermal wind equation geostrophic_eastward_sea_water_velocity
vgo (m s-1): geostrophic meridional velocity from thermal wind equation geostrophic_northward_sea_water_velocity
mlotst (meter): mixed layer depth from min of temperature and density threshold equivalent to a 0.2°C decrease ocean_mixed_layer_thickness
Dataset: **cmems_obs-mob_glo_phy_mynrt_0.125deg-climatology-uncertainty_P1M-m** Variables name in the NetCDF file and Unit: Long_name & Standard_name

to_uncertainty (degree Celsius)**:** sea temperature monthly uncertainty sea_water_temperature_uncertainty
so_uncertainty (practical salinity unit)**:** sea salinity monthly uncertainty sea_water_salinity_uncertainty

d) Catalogue updating

The daily NRT dataset (cmems_obs-mob_glo_phy_nrt_0.125deg_P1D-m) is produced daily and made available each day for Day-6 analysis.
The cmems_obs-mob_glo_phy_nrt_0.125deg_P1M-m provides the monthly average of the daily dataset. NRT monthly average (cmems_obs-mob_glo_phy_nrt_0.125deg_P1M-m) is provided the 6th of each following month.
The multi-year datasets (cmems_obs-mob_glo_phy_my_0.125deg_P1D-m, cmems_obs-mob_glo_phy_my_0.125deg_P1M-m) extension is carried out once a year.
Static uncertainties (cmems_obs-mob_glo_phy_mynrt_0.125deg-climatology-uncertainty_P1M-m) are updated when major changes are made to the product (full reprocessing).

e) Time averaging

Dataset frequency	Time averaging
montlhy	monthly means over the calendar month (first to last day of the month)
daily	daily means over a day (midnight to midnight, centered at noon).

f) Product use and impact

The product combined different observations datasets, satellites and in situ, with a statistical approach. In this sense, it is complementary to numerical equations-based oceanic models (delivered by the MFCs of the Copernicus Marine Service). It delivers 3D fields of temperature, salinity, geostrophic currents and mixed layer depth. From November 2025, the datasets are provided at a daily temporal resolution and 1/8° spatial resolution. The product can be used for numerous scientific analyses of the physical ocean and for validation purposes. It represents the mesoscale measured by satellites (sea level, sea surface temperature, sea surface salinity) and this guarantees a good location of the mesoscale features, and it is close to the in situ observations of temperature and salinity. The validation of the geostrophic velocities also indicates good agreement with the observations.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 files nomenclature is:

cmems_obs-mob_glo_phy_<type>_0.125deg_P1D-m:

cmems_obs-mob_glo_phy_<type>_0.125deg_P1D-m_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc

with type={nrt; my}

YYYYMMDDThhmmZ: date of the product;

PYYYYMMDDThhmmZ: date of the production

cmems_obs-mob_glo_phy_<type>_0.125deg_P1M-m:

cmems_obs-mob_glo_phy_<type>_0.125deg_P1M-m_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc

with type={nrt; my};

YYYYMM15ThhmmZ: date of the product;

PYYYYMMDDThhmmZ: date of the production

cmems_obs-mob_glo_phy_mynrt_0.125deg-climatology-uncertainty_P1M-m:

dataset-armor-3d-monthly_climatology_uncertainty_P1M-m<xx>.nc

with xx=month number from 01 to 12

4.1 FILE SIZE

DATASET NAME	FILE NAME	DIMENSION [MB]
cmems_obs-mob_glo_phy_<type>_0.125deg_<resolution>-m type={nrt;my} resolution={P1D; P1M}	cmems_obs-mob_glo_phy_<type>_0.125deg_<resolution>-m_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc type={nrt;my} resolution={ P1D; P1M }	~641

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex,** for each dataset.

5) REFERENCES

Quality Information Document (QUID) :
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-MOB-QUID-015-012.pdf

6) ANNEX

One example header is given below for **cmems_obs-mob_glo_phy_my_0.125deg_P1D-m**. The three other datasets (my monthly, nrt monthly and daily) are similar.

```
netcdf cmems_obs-mob_glo_phy_my_0.125deg_P1D-
m_20181231T1200Z_P20250409T0313Z {
dimensions:
        time = UNLIMITED ; // (1 currently)
        depth = 50 ;
        latitude = 1378 ;
        longitude = 2880 ;
variables:
        int time(time) ;
                time:units = "hours since 1950-01-01" ;
                time:axis = "T" ;
                time:long_name = "time" ;
                time:standard_name = "time" ;
                time:calendar = "gregorian" ;
        short depth(depth) ;
                depth:axis = "Z" ;
                depth:long_name = "depth" ;
                depth:positive = "down" ;
                depth:standard_name = "depth" ;
                depth:unit_long = "meter" ;
                depth:units = "m" ;
        float latitude(latitude) ;
                latitude:axis = "Y" ;
                latitude:long_name = "latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:step = 0.125f ;
                latitude:unit_long = "degrees" ;
                latitude:units = "degrees" ;
        float longitude(longitude) ;
                longitude:axis = "X" ;
                longitude:long_name = "longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:step = 0.125f ;
                longitude:unit_long = "degrees" ;
                longitude:units = "degrees" ;
        short so(time, depth, latitude, longitude) ;
                so:_FillValue = 32767s ;
                so:add_offset = 20. ;
                so:long_name = "salinity" ;
                so:scale_factor = 0.001 ;
                so:standard_name = "sea_water_salinity" ;
                so:unit_long = "practical salinity unit" ;
                so:units = "0.001" ;
        short to(time, depth, latitude, longitude) ;
                to:_FillValue = 32767s ;
                to:add_offset = 20. ;
                to:long_name = "temperature" ;
                to:scale_factor = 0.001 ;
                to:standard_name = "sea_water_temperature" ;
```

```
                to:unit_long = "degree Celsius" ;
                to:units = "degrees_C" ;
        short ugo(time, depth, latitude, longitude) ;
                ugo:_FillValue = 32767s ;
                ugo:long_name = "geostrophic zonal velocity from thermal wind equation" ;
                ugo:scale_factor = 0.001 ;
                ugo:standard_name = "geostrophic_eastward_sea_water_velocity" ;
                ugo:unit_long = "meter per second" ;
                ugo:units = "m s-1" ;
                ugo:valid_range = -4000s, 4000s ;
        short vgo(time, depth, latitude, longitude) ;
                vgo:_FillValue = 32767s ;
                vgo:long_name = "geostrophic meridional velocity from thermal wind equation" ;
                vgo:scale_factor = 0.001 ;
                vgo:standard_name = "geostrophic_northward_sea_water_velocity" ;
                vgo:unit_long = "meter per second" ;
                vgo:units = "m s-1" ;
                vgo:valid_range = -4000s, 4000s ;
        short zo(time, depth, latitude, longitude) ;
                zo:_FillValue = 32767s ;
                zo:long_name = "absolute height" ;
                zo:scale_factor = 0.001 ;
                zo:standard_name = "geopotential_height" ;
                zo:unit_long = "meter" ;
                zo:units = "m" ;
                zo:valid_range = -20000s, 20000s ;
        short mlotst(time, latitude, longitude) ;
                mlotst:_FillValue = 32767s ;
                mlotst:scale_factor = 0.1 ;
                mlotst:unit_long = "meter" ;
                mlotst:add_offset = 2500. ;
                mlotst:long_name = "Mixed layer depth from min of temperature and density
threshold equivalent to a 0.2C decrease" ;
                mlotst:standard_name = "ocean_mixed_layer_thickness" ;
                mlotst:units = "m" ;

// global attributes:
                :description = "ARMOR3D REP Copernicus Marine Service November 2025
Release" ;
                :title = "ARMOR3D REP - TSHUVMld Global Ocean Observation-based
Product" ;
                :Conventions = "CF-1.0" ;
                :institution = "CLS" ;
                :domain_name = "GLO" ;
                :history = "2025-04-09 03:14:30 ARMOR3D REP - TSHUV Global Ocean
Observation-based Product netCDF creation" ;
                :lon_min = -179.9375 ;
                :lon_max = 179.9375 ;
                :lat_min = -82.1875 ;
                :lat_max = 89.9375 ;
}
```

One example header is given below for **cmems_obs-mob_glo_phy_mynrt_0.125deg-climatology-uncertainty_P1M-m:**

```
netcdf dataset-armor-3d-monthly_climatology_uncertainty_P1M-m_01 {
```

```
dimensions:
        time = 1 ;
        depth = 41 ;
        latitude = 1378 ;
        longitude = 2880 ;
variables:
        string time(time) ;
                time:long_name = "Monthly climatology of month 1" ;
        double to_uncertainty(time, depth, latitude, longitude) ;
                to_uncertainty:_FillValue = 32767. ;
                to_uncertainty:units = "degree Celsius" ;
                to_uncertainty:long_name = "sea temperature monthly uncertainty" ;
                to_uncertainty:standard_name = "sea_water_temperature_uncertainty" ;
        double so_uncertainty(time, depth, latitude, longitude) ;
                so_uncertainty:_FillValue = 32767. ;
                so_uncertainty:units = "practical salinity unit" ;
                so_uncertainty:long_name = "sea salinity monthly uncertainty" ;
                so_uncertainty:standard_name = "sea_water_salinity_uncertainty" ;
        float depth(depth) ;
                depth:_FillValue = NaNf ;
                depth:axis = "Z" ;
                depth:long_name = "depth" ;
                depth:positive = "down" ;
                depth:standard_name = "depth" ;
                depth:unit_long = "meter" ;
                depth:units = "m" ;
        float latitude(latitude) ;
                latitude:_FillValue = NaNf ;
                latitude:axis = "Y" ;
                latitude:long_name = "latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:step = "0.125f" ;
                latitude:unit_long = "degrees_north" ;
                latitude:units = "degrees_north" ;
        float longitude(longitude) ;
                longitude:_FillValue = NaNf ;
                longitude:axis = "X" ;
                longitude:long_name = "longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:step = "0.125f" ;
                longitude:unit_long = "degrees_north" ;
                longitude:units = "degrees_east" ;

// global attributes:
                :description = "ARMOR3D REP Copernicus Marine Service November 2024
Release" ;
                :title = "ARMOR3D - TS Global Ocean Observation-based Product
Uncertainties" ;
                :Conventions = "CF-1.0" ;
                :institution = "CLS" ;
                :domain_name = "GLO" ;
                :history = "2024-10-15 15:04:52 ARMOR3D - TS Global Ocean Observation-
based Product netCDF Uncertainties creation" ;
}
```